

Gate Smashers AI & ML Playlist Exam Mapping Guide

Course Pattern: TE IT - 2024 Pattern (SPPU)
Document Type: Exam Playlist Mapping

Academic Term: Semester V (2026)
Course Code: PCC-302-IT

Introduction and Purpose

This document provides an exam-focused, structured mapping of the Gate Smashers Artificial Intelligence and Machine Learning YouTube playlists to the official SPPU TE IT (2024 Pattern) Artificial Intelligence & Machine Learning Syllabus (PCC-302-IT). It classifies 102 core lectures from both official playlists into the five syllabus units.

- **Artificial Intelligence Playlist:** Gate Smashers AI Playlist ↗
- **Machine Learning Playlist:** Gate Smashers ML Playlist ↗

Exam Preparation Note: Lectures highlighted in **bold** represent high-priority exam topics explicitly tested in SPPU end-semester exams and fully aligned with the unit-wise theory notes (such as Turing Test, A* Search, Hill Climbing, Propositional/Predicate Logic, Linear/Logistic Regression, Decision Trees, SVM, k-Means, PCA, and Neural Networks). Focus on these lectures first for maximum marks!

Quick Statistics:

- **Total Mapped Playlist Videos:** 102 (Total Duration: 16h 44m)
- **Unit 1 (Introduction to AI and Intelligent Agents):** 11 Videos (1h 28m)
- **Unit 2 (Problem Solving, Search & Knowledge Representation):** 26 Videos (4h 34m)
- **Unit 3 (Machine Learning Foundations & Supervised Learning):** 21 Videos (3h 36m)
- **Unit 4 (Unsupervised Learning, Ensemble Methods & Preprocessing):** 20 Videos (2h 46m)
- **Unit 5 (Neural Networks, Deep Learning & Advanced AI):** 24 Videos (4h 18m)

Unit 1: Introduction to AI and Intelligent Agents (1h 28m)

Syllabus Topics Covered: Definitions of AI, Turing Test, History & Evolution, ANI/AGI/ASI, AI vs ML vs DL vs DS, Intelligent Agents (Sensors, Actuators, Percept sequence), Rationality, PEAS Framework, 7 Environment dimensions, 5 Agent Architectures, Applications & Ethics.

Lecture Topic / Title	Duration	Watch Link
Lec-1: Artificial Intelligence Syllabus Discussion and Analysis for NTA UGC NET	11:26	Play ↗
Lec-2: What's the FUTURE of Artificial Intelligence in 2025	4:30	Play ↗
Lec-3: What is Artificial Intelligence I Learn AI with Real Life Examples I Can Machine Think??	8:31	Play ↗
Lec-28: Introduction to Intelligent Agents and their types with Example in Artificial Intelligence	11:10	Play ↗
Lec-29: Simple Reflex Agent in Artificial Intelligence with Example I Artificial Intelligence	9:04	Play ↗

Lecture Topic / Title	Duration	Watch Link
Lec-30: Model Based Reflex Agent in Artificial Intelligence in HINDI with Real Life Examples	7:15	Play ↗
Lec-31: Goal Based Agents in Artificial Intelligence with real life examples in HINDI	5:28	Play ↗
Lec-32: Utility Based Agents in Artificial Intelligence in Hindi with real life examples	5:35	Play ↗
Lec-37: Supervised, Unsupervised and Reinforcement Learning in Artificial Intelligence in Hindi	9:28	Play ↗
Lec-1: Introduction to Data Science & ML Roadmap to Learn Data Science & ML	8:24	Play ↗
Lec-23: Supervised vs Unsupervised learning with real life example	7:31	Play ↗

Unit 2: Problem Solving, Search and Knowledge Representation (4h 34m)

Syllabus Topics Covered: Problem Formulation, State Space, 8-Puzzle, Uninformed Search (BFS, DFS, UCS), Informed Search (Heuristics, Greedy Best-First, A Search, Admissibility & Consistency), Local Search & Hill Climbing (Local Maxima, Ridges, Plateaux), Knowledge Representation, Propositional & Predicate Logic (Quantifiers, WFF), Forward & Backward Chaining, Expert Systems.*

Lecture Topic / Title	Duration	Watch Link
Lec-4: What is State Space Search Introduction to Problem Solving in Artificial Intelligence	10:47	Play ↗
Lec-5: Uninformed Vs Informed Search in Artificial Intelligence with Example	8:17	Play ↗
Lec-6: Informed vs Uninformed vs Adversarial Search With Examples Artificial Intelligence	6:23	Play ↗
Lec-7: Breadth First Search (BFS) with example Uninformed Search Artificial Intelligence	12:57	Play ↗
Lec-8: Depth First Search (DFS) with example Uninformed Search Artificial Intelligence	9:13	Play ↗
Lec-9: Time & Space Complexity of BFS and DFS Artificial Intelligence	7:58	Play ↗
Lec-10: Bidirectional Search Algorithm in Artificial Intelligence in Hindi with Real Life Examples	8:47	Play ↗
Lec-11: Best First Search Algorithm How it Works All Imp Points(Pros & Cons)	14:41	Play ↗
Lec-12: Depth-Limited Search (DLS) Explained Uninformed Search in AI	13:11	Play ↗
Lec-13: Iterative Deepening Depth-First Search (IDDFS) Artificial Intelligence	5:11	Play ↗
Lec-14: 8-Puzzle Problem in Artificial Intelligence without Heuristic All Imp Points Must Watch	11:05	Play ↗

Lecture Topic / Title	Duration	Watch Link
Lec-15: What is Heuristic in AI Why we use Heuristic How to Calculate Heuristic Must Watch	12:57	Play ↗
Lec-16: How to Solve 8-Puzzle Problem with Heuristic(Informed Search) in Artificial Intelligence	11:07	Play ↗
Lec-17: Generate and Test Search in Artificial Intelligence with Examples Heuristic Search	7:58	Play ↗
Lec-18: Beam Search Algorithm All Imp Points Heuristic Search Techniques	12:02	Play ↗
Lec-19: Hill Climbing Algorithm in Artificial Intelligence with Real Life Examples Heuristic Search	10:14	Play ↗
Lec-20: A* algorithm in AI (artificial intelligence) in HINDI A* algorithm with example	14:34	Play ↗
Lec-21: How to Proof A* Admissible Underestimation & Overestimation of A* in Hindi	12:25	Play ↗
Lec-22: AO* algorithm in AI (artificial intelligence) in HINDI AO* algorithm with example	13:24	Play ↗
Lec-26: Knowledge Representation and Reasoning Logic, Semantic Net, Frames etc.	7:44	Play ↗
Lec-27: Propositional Logic in Artificial Intelligence Knowledge Representation All Imp Points	12:20	Play ↗
Lec-43: Propositional Logic Artificial Intelligence	11:47	Play ↗
Lec-44: Predicate Logic Artificial Intelligence	10:11	Play ↗
Lec-45: How to write First order/Predicate logic Artificial Intelligence	9:24	Play ↗
Lec-46: Negation of Quantifiers Predicate Logic Logic with Certainty Artificial Intelligence	7:05	Play ↗
Lec-47: Questions on Propositional & Predicate logic How to Solve Problems in Logic	12:46	Play ↗

Unit 3: Machine Learning Foundations & Supervised Learning (3h 36m)

Syllabus Topics Covered: Supervised vs Unsupervised Learning, Regression Models (Linear, Multiple, Ridge, Lasso, Logistic), Classification Algorithms (kNN, Naive Bayes, Decision Trees ID3, SVMs), Model Evaluation Metrics (MSE, Bias-Variance Tradeoff, Overfitting/Underfitting), Cross-Validation (K-Fold, LOOCV).

Lecture Topic / Title	Duration	Watch Link
Lec-2: Supervised Learning Algorithms Machine Learning	8:50	Play ↗
Lec-3: Introduction to Regression with Real Life Examples	7:19	Play ↗
Lec-4: Linear Regression  with Real life examples & Calculations Easiest Explanation	11:01	Play ↗

Lecture Topic / Title	Duration	Watch Link
Lec-5: Logistic Regression with Simplest & Easiest Example Machine Learning	10:01	Play ↗
Lec-6: Linear Regression Vs. Logistic Regression Supervised Learning Machine Learning	4:37	Play ↗
Lec-7: kNN Classification with Real Life Example Movie Imdb Example Supervised Learning	10:13	Play ↗
Lec-8: Naive Bayes Classification Full Explanation with examples Supervised Learning	13:31	Play ↗
Lec-9: Introduction to Decision Tree 🌳 with Real life examples	6:07	Play ↗
Lec-10: Decision Tree 🌳 ID3 Algorithm with Example & Calculations	16:38	Play ↗
Lec-19: kNN for Classified & Regression with Easiest Explanation Machine Learning 🧠💡	7:21	Play ↗
Lec-26: Cross Validation in Machine Learning with Examples	6:51	Play ↗
Lec-28: kNN(k Nearest Neighbour) Numerical Example Supervised Learning Machine Learning	9:09	Play ↗
Lec-29: Decision Tree 🌳 Example Calculate Entropy, Information Gain Supervised Learning	6:57	Play ↗
Lec-37: Ridge and Lasso Regression Machine Learning	14:10	Play ↗
Lec-38: Mean Squared Error (MSE) Machine learning	9:53	Play ↗
Lec-39: Multiple Linear Regression (MLR) Machine Learning	12:48	Play ↗
Lec-40: Support Vector Machines (SVMs) Machine Learning	10:23	Play ↗
Lec-41: Numerical Explanation on SVM How Support Vector Machine Algorithm Works	16:07	Play ↗
Lec-43: Bias & Variance Tradeoff Explained: How to Fix Overfitting & Underfitting?	14:44	Play ↗
Lec-44: K-Fold Cross Validation in Machine Learning	9:52	Play ↗
Lec-45: Leave-One-Out Cross Validation (LOOCV) Explained with Example Machine Learning	9:36	Play ↗

Unit 4: Unsupervised Learning, Ensemble Methods & Preprocessing (2h 46m)

Syllabus Topics Covered: Clustering Algorithms (k-Means, k-Medoids, Agglomerative & Divisive Hierarchical Clustering, Single/Complete Linkage), Ensemble Learning (Bagging, Random Forest, Boosting, Stacking), Feature Engineering & Dimensionality Reduction (PCA, LDA, Pearson Correlation), Data Preprocessing (Missing value imputation, kNN imputation, Fit/Transform).

Lecture Topic / Title	Duration	Watch Link
Lec-12: Introduction to Ensemble Learning with Real Life Examples Machine Learning 🧠	5:58	Play ↗

Lecture Topic / Title	Duration	Watch Link
Lec-13: K-mean Clustering with Numerical Example Unsupervised Learning Machine Learning 🧠 ♂ 🤖	7:51	Play ↗
Lec-14: Hierarchical Clustering Agglomerative vs Divisive with examples	6:06	Play ↗
Lec-15: Single Linkage Clustering Agglomerative Clustering Hierarchical Clustering	6:16	Play ↗
Lec-16: Complete Linkage Clustering with Example Clustering in Unsupervised Learning ML	9:05	Play ↗
Lec-17: K-medoids Clustering with Numerical Example Machine Learning	11:53	Play ↗
Lec-18: Random Forest 🌲 in Machine Learning 🤖 🤖	8:33	Play ↗
Lec-22: Bagging/Bootstrap Aggregating in Machine Learning with examples	4:56	Play ↗
Lec-24: How Weights are Increased in Boosting Ensemble Learning	6:48	Play ↗
Lec-25: BAGGING vs. BOOSTING vs STACKING in Ensemble Learning Machine Learning	6:22	Play ↗
Lec-27: Pearson's Correlation Coefficient Supervised Learning Data Science & Machine Learning	7:38	Play ↗
Lec-30: Single Linkage Clustering Example Unsupervised Learning Machine Learning	6:52	Play ↗
Lec-32: What is Data Preprocessing & Data Cleaning Various Techniques with Example	5:53	Play ↗
Lec-33: How to Deal with Missing Values in DataSet Data Preprocessing & Data Cleaning	9:27	Play ↗
Lec-34: kNN Imputation with Examples Data Preprocessing and Data Cleaning ✍️	7:51	Play ↗
Lec-35: Fit() & Transform() Method Data Preprocessing Machine Learning	6:36	Play ↗
Lec-36: Feature Extraction in Data preprocessing Machine Learning	9:21	Play ↗
Lec-42: Linear Discriminant Analysis (LDA) Machine Learning	13:21	Play ↗
Lec-46: Principal Component Analysis (PCA) Explained Machine Learning	14:06	Play ↗
Lec-47: How to update cost in K-Medoid Clustering Machine Learning	12:05	Play ↗

Unit 5: Neural Networks, Deep Learning & Advanced AI (4h 18m)

Syllabus Topics Covered: Artificial Neural Networks (Single Layer Perceptron, Multilayer Perceptron MLP), Fuzzy Logic & NLP, Constraint Satisfaction Problems (CSP, Backtracking, Branch & Bound, 0/1 Knapsack), Probabilistic

Reasoning under Uncertainty (Bayesian Networks, Likelihood Weight Sampling, Rejection Sampling, Markov Models, Hidden Markov Models HMM), Modern LLMs & Transformer Parameters.

Lecture Topic / Title	Duration	Watch Link
Lec-33: Fuzzy Logic in Artificial Intelligence with Example I Artificial Intelligence	13:03	Play ↗
Lec-34: Various Operations in Fuzzy Logic with Example I Union, Intersection, Complement etc.	13:02	Play ↗
Lec-35: Introduction to Neural Networks with Example in HINDI I Artificial Intelligence	11:20	Play ↗
Lec-36: Natural Language Processing in Artificial Intelligence in Hindi I NLP with Demo and Examples	17:18	Play ↗
Lec-38: Genetic Algorithm in Artificial Intelligence I Simplest Explanation with real life examples	12:24	Play ↗
Lec-39: What is Constraint Satisfaction I Constraint Satisfaction Problem (CSP) in AI with Example	8:11	Play ↗
Lec-40: How Constraint Satisfaction Algorithm Works I Explained with Interesting Example I AI	10:40	Play ↗
Lec-41: Branch & Bound Algorithm with Example I Easiest Explanation of B&B with example	13:13	Play ↗
Lec-42 : 0/1 Knapsack using Branch and Bound with example	11:20	Play ↗
Lec-48: Bayes Theorem & Total Probability with Examples	7:16	Play ↗
Lec-49 : Network with Examples I Easiest Explanation	9:54	Play ↗
Lec-50: Question on Bayesian Network I Artificial Intelligence	7:47	Play ↗
Lec-51: Bayesian Network Numerical Example #gate2025preparation	7:10	Play ↗
Lec-52: Probabilistic Inference I Sampling I Artificial Intelligence	8:06	Play ↗
Lec-53: Likelihood Weight Sampling I Inference through Sampling I Uncertainty in Artificial	9:08	Play ↗
Lec-54: Question on Likelihood Sampling I Probabilistic Inference I Sampling I Artificial	8:07	Play ↗
Lec-55: Rejection Sampling I Probabilistic Inference I Sampling I Artificial Intelligence	6:29	Play ↗
Lec-56: Reasoning under Uncertainty in Artificial Intelligence	8:28	Play ↗
Lec-57: Uncertainty probabilistic inference (Markov Model) I Artificial Intelligence	14:15	Play ↗
Lec-58: Introduction to Hidden Markov Model I Artificial Intelligence	13:49	Play ↗
Lec-31: Token & Parameters in LLama3 META Models I 8B & 70B Parameters Model I GPT model	7:09	Play ↗
Lec-48: Perceptron Learning in ANN I Single Layer Perceptron Model	15:04	Play ↗
Lec-49: What is Multilayer Perceptron (MLP)? I How It Works in Machine Learning	12:56	Play ↗
Lec-50: Single Layer Neural Network I Machine Learning	12:03	Play ↗